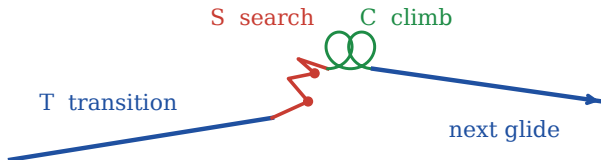


How directional persistence shapes the Hurst-exponent crossover of thermal soaring flights

A cycle-based CTRW for cross-country soaring

Target: the observed $\text{MSD} \propto t^{1.76}$ and its collapse across aircraft classes.



We choose **Lomax** durations for **T** and **S**, as they appear to describe Vilpellet et al.'s empirical survival curves:

$$P(\tau^X > t) = \left(1 + \frac{t}{\tau_0^X}\right)^{-\mu_X}, \quad X = T, S.$$

We model **C** with an **exponential** duration:

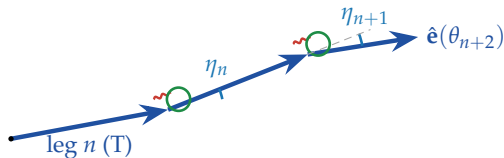
$$P(\tau^C > t) = e^{-t/(110\text{s})}.$$

The chosen parameters have $\mu_T, \mu_S > 2$: the cycle clock has finite mean and variance.

The memory between cycles resides in the glide direction.

Trajectory: schematic. Independent phase durations, i.i.d. across cycles. Empirical inputs: Vilpellet, Darmon and Benzaquen (2026), arXiv:2601.01293.

$$\theta_{n+1} = \theta_n + \eta_n \pmod{2\pi}, \quad \eta_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\theta^2)$$

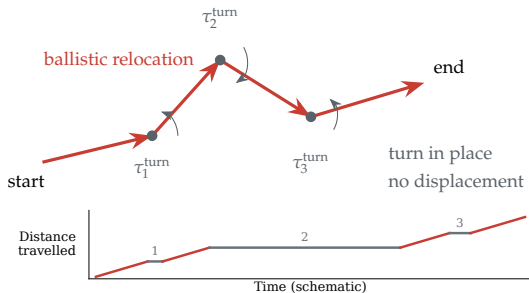


$$C(k) = \langle \hat{\mathbf{e}}(\theta_{n+k}) \cdot \hat{\mathbf{e}}(\theta_n) \rangle = e^{-k\sigma_\theta^2/2} = \rho^k, \quad n_c = \frac{2}{\sigma_\theta^2}.$$

k counts successive glides, after segmentation. It is a cycle lag, not a time lag in seconds.

At fixed speed, $\mathbf{v}_n = v_{xy} \hat{\mathbf{e}}(\theta_n)$, so $C(k) = \langle \mathbf{v}_{n+k} \cdot \mathbf{v}_n \rangle / v_{xy}^2$.

θ_0 is uniform. Glide length $\ell_n = v_{xy} \tau_n^T$ is independent of headings. One glide is ballistic: $\delta_T^2(\Delta) = v_{xy}^2 \Delta^2$. S and C have independent, isotropic local angles.



MSD within a search episode

$$\delta_S^2(\Delta) \simeq \begin{cases} u_S^2 \Delta^2 & \text{short lag,} \\ K_S \Delta^{\alpha_S} & \text{after decorrelation.} \end{cases}$$

The renewal clock sets the subdiffusive exponent.

Schematic path and clock. Curved arrows show turns at fixed position. Turning update: $\psi_{j+1} = \psi_j + \epsilon_j \Omega_S \tau_j^{\text{turn}}$, with $P(\epsilon_j = \pm 1) = 1/2$. An independent Lomax duration truncates the episode.

Short ballistic relocations

$$\tau^b \sim \text{Exp}(1/\tau_b), \quad \tau_b = 10 \text{ s.}$$

Mittag-Leffler turning waits

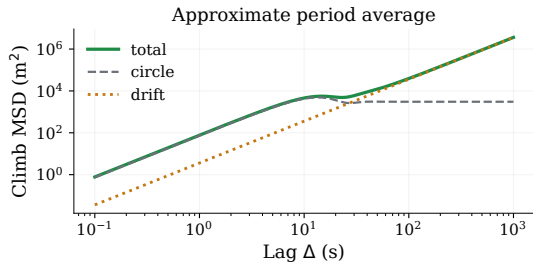
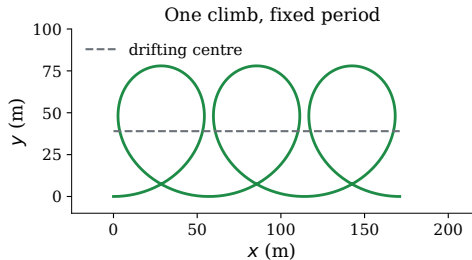
$$P(\tau^{\text{turn}} > t) = E_{\alpha_S}[-(t/\tau_{\text{turn}})^{\alpha_S}].$$

No net displacement during a wait.

$$\alpha_S = 0.6, 0.6, 0.4$$

for para., hang gliders, sailplanes.

$$\mathbf{x}_C(t) = r_0(\cos(\omega t + \phi), \sin(\omega t + \phi)) + \mathbf{v}_{\text{drift}}t.$$



$$\delta_C^2(\Delta) \simeq \underbrace{2r_0^2[1 - e^{-\sigma_\omega^2 \Delta^2/2} \cos(\bar{\omega} \Delta)]}_{\text{circle + period dispersion}} + \underbrace{v_{\text{drift}}^2 \Delta^2}_{\text{drift}}.$$

$T_{\text{turn}} \sim \mathcal{N}(\bar{T}, \sigma_T^2)$, clipped at $0.2\bar{T}$. Orbital phase and drift direction are independent and uniform.

Illustration: paraglider parameters. $\bar{\omega} = 2\pi/\bar{T}$, $\sigma_\omega \simeq \bar{\omega}\sigma_T/\bar{T}$. The closed form is a first-order period-dispersion approximation. The exact period average matches Monte Carlo. Short-lag errors can reach a factor of two for sailplanes.

$$\mathbf{x}_N = \underbrace{\sum_{n=1}^N \ell_n \hat{\mathbf{e}}(\theta_n)}_{\text{transition}} + \underbrace{\sum_{n=1}^N \Delta \mathbf{x}_n^S}_{\text{search}} + \underbrace{\sum_{n=1}^N \Delta \mathbf{x}_n^C}_{\text{climb}}.$$

MSD of transition glides

$$M_T(N) = \underbrace{N \text{Var}(\ell)}_{\text{diffusive}} + \underbrace{\langle \ell \rangle^2}_{\text{coherent}} G_N.$$

Search and climb add linearly

$$M_S(N) + M_C(N) = N(\Sigma_S + \Sigma_C).$$

$$G_N \equiv N + 2 \sum_{k=1}^{N-1} (N-k) C(k), \quad C(k) = \rho^k, \quad \rho = e^{-\sigma_\theta^2/2}.$$

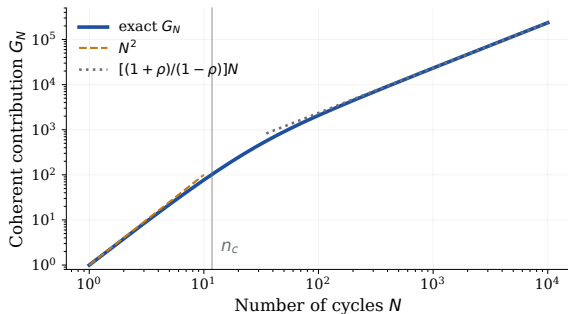
$$\boxed{M_N = \mathcal{A} G_N + \mathcal{B} N}, \quad \mathcal{A} = \langle \ell \rangle^2, \quad \mathcal{B} = \text{Var}(\ell) + \Sigma_S + \Sigma_C.$$

S and C are isotropic and independent across episodes and from T. Their cross terms vanish.

Local search subdiffusion becomes a diffusive contribution across cycles.

$\Sigma_X = \langle |\Delta \mathbf{x}_n^X|^2 \rangle < \infty$ averages complete episodes over their durations. For the fitted inputs, $\text{Var}(\ell)$ supplies more than 97% of $\mathcal{B}$.

$$G_N = N \frac{1 + \rho}{1 - \rho} - \frac{2\rho(1 - \rho^N)}{(1 - \rho)^2}, \quad \rho = e^{-\sigma_\theta^2/2}.$$



Few cycles: $N \ll n_c$

$$G_N \simeq N^2.$$

Aligned legs add coherently.

Many cycles: $N \gg n_c$

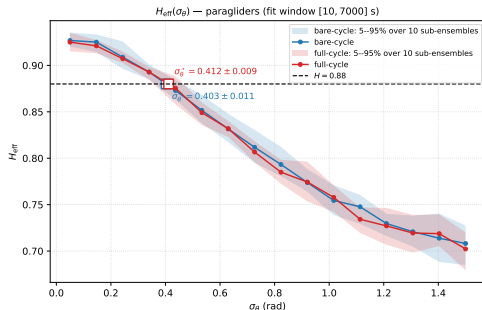
$$G_N \simeq \frac{1 + \rho}{1 - \rho} N.$$

The long-time limit has $H = 1/2$.

Finite windows can display $H \simeq 0.88$ during the crossover.

Exact in cycle count. $n_c = 2/\sigma_\theta^2$ and $t_c \simeq n_c \langle T_{\text{cyc}} \rangle$. Illustration: $\sigma_\theta = 0.412$ rad. The total MSD also contains $\mathcal{B}N$; ballistic scaling requires the coherent term to dominate.

Calibration target: $H_{\text{fit}} = 0.88$ over lags 10–7000 s.



	σ_θ^* (rad)	n_c
Para.	$0.412 \pm .009$	11.8
Hang gl.	$0.242 \pm .005$	34.2
Sailpl.	$0.556 \pm .014$	6.5

$t_c \simeq 4.9, 15.1, 3.1$ ks.

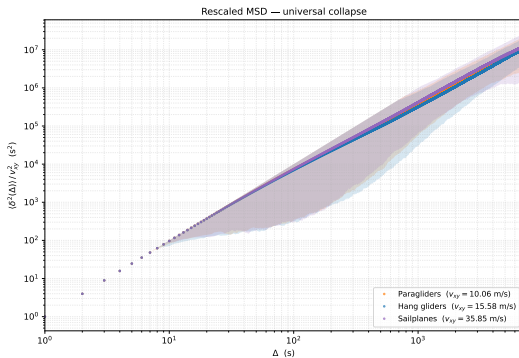
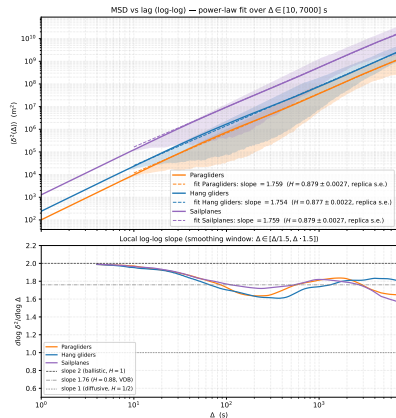
Only σ_θ is fitted per class.

Other inputs come from empirical statistics and figures.

The value $H \simeq 0.88$ is a calibration target.

16 values of σ_θ in $[0.05, 1.5]$ rad. Per value: 2,000 flights of 15,000 s, sampled every second. Log–log OLS on integer lags. Quoted errors: replica standard errors. Figure bands: 5–95% spread.

Fresh ensemble: 2,000 flights per class, 15,000 s each. MSD measured from each flight's origin.

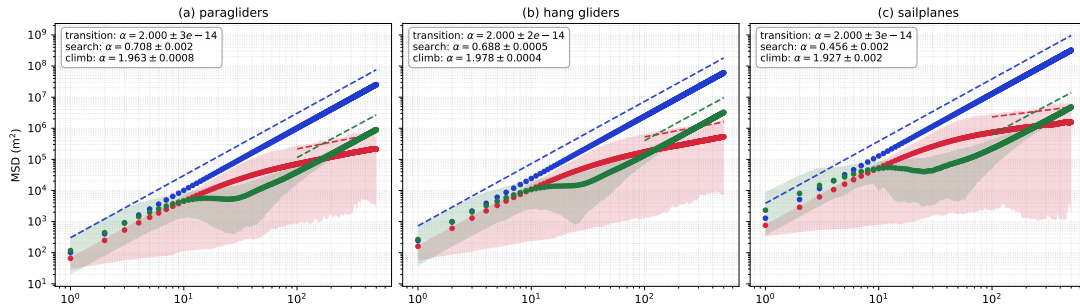


Dividing by v_{xy}^2 collapses the curves.

$$H_{\text{fit}} = 0.879 \pm .003, 0.877 \pm .002, 0.879 \pm .003.$$

Class order: para., hang gliders, sailplanes. Both $\mathcal{A}$ and the dominant $\text{Var}(\ell)$ scale as v_{xy}^2 . Bands show 5–95% spread of individual squared displacements, not fit uncertainty.

Episode-origin MSD, conditioned on the episode lasting at least lag Δ (seconds).



Transition

Ballistic: exponent 2.

Search

Subdiffusive: 0.708,
0.688, 0.456.

Climb

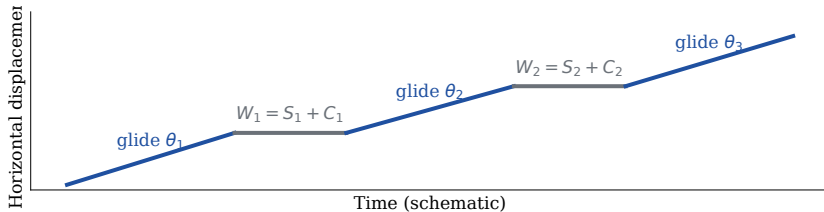
Dip, then drift:
1.96, 1.98, 1.93.

No individual phase gives exponent 1.76.

Monte Carlo: 2,000 flights of 50 cycles per class. Search inputs $\alpha_S = 0.6, 0.6, 0.4$; finite episodes limit the asymptotic regime. Exponents follow panel order and plotted fit annotations. Bands: 5–95% episode spread.

Merge search and climb into a single waiting time:

$$W_n = \tau_n^S + \tau_n^C, \quad T_{\text{cyc},n} = \tau_n^T + W_n.$$



Preserve the clock and memory

Glides are ballistic. Position is fixed during W_n . The next glide retains angular memory.

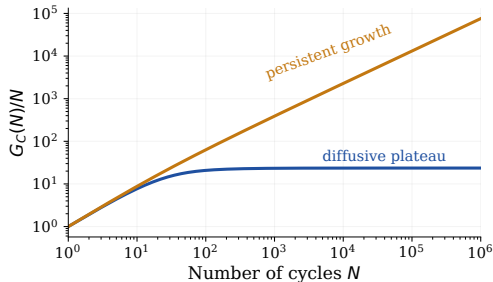
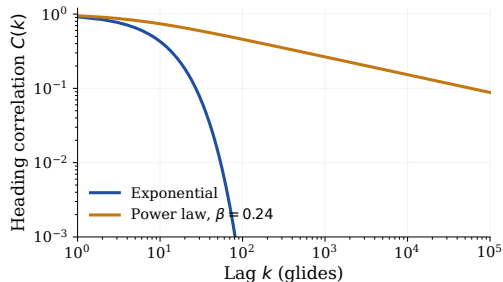
Remove local displacements

The full MSD differs by $N(\Sigma_S + \Sigma_C)$ at cycle boundaries.

A large-scale approximation with the same angular transport mechanism.

W follows the convolution of the search Lomax law and the climb exponential law. Its mean and variance remain finite. This simplification removes S/C phase MSDs and their short-time structure. Plot: schematic horizontal coordinate.

$$M_N = (\mathcal{A} + \mathcal{B})N + 2\mathcal{A} \sum_{k=1}^{N-1} (N-k)C(k).$$



$$C(k) \sim ck^{-\beta}, c > 0, 0 < \beta < 1$$

$$M_N \propto N^{2-\beta}, \quad H = 1 - \frac{\beta}{2}.$$

Boundary and diffusive cases

$$\beta = 1: M_N \sim N \log N.$$

$$\beta > 1 \text{ or exponential: } M_N \propto N.$$